MIL-C-52153B
25 August 1982
SUPERSEDING
MIL-C-52153A
2 November 1973

MILITARY SPECIFICATION

COMPRESSOR UNIT, RECIPROCATING, POWER-DRIVEN: AIR; 5-CFM, 175-PSI, FOR TESTING MISSILE PLUMBING SYSTEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 5-cubic-foot-per-minute (cfm), 175-pound-per-square-inch-gage (psig) compressor equipped for testing missile propulsion plumbing and associated assemblies.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-G-91 - Gages, Tire Pressure, Self Contained and Inflator-Gage, Pneumatic Tire.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4310

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MILITARY

MIL-C-3600	- Compressors, Rotary, Power-Driven; and Compressors, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-C-13874	- Compressors, Reciprocating and Rotary, Power-Driven, Air.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1474	- Noise Limits for Army Materiel.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION

National Electrical Code (No. 70).

(Application for copies should be addressed to the National Fire Protection Association, 60 Battermarch Street, Boston, MA 02110.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The compressor unit (hereinafter called "compressor") for testing missile propulsion plumbing shall consist of an air compressor conforming to MIL-C-13874, type I, class 3, size 5 cfm at 175 psig, with additions and exceptions to MIL-G-13874 as specified herein. The air receiver tank capacity shall be 30 gallons. The electric motor shall operate on a 115/230-volt, single-phase, 60 Hertz, power supply.

3.2 First articles.

3.2.1 Preproduction model. The contractor shall furnish one or more compressors as specified (see 6.2), for examination and test within the time

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frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce compressors that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests.

3.3 Cable. The power supply cable shall conform to National Electrical Code, ANSI/NFPA70 including grounding. The cord shall be 3-conductor, with the conductor size not smaller than No. 10 American Wire Gage (AWG), and shall be not less than 50 feet long. Unless otherwise specified (see 6.2), the cable ends shall not be terminated.

3.4 Handle and storage rack. A combination handle and storage rack, similar to that shown on figure 1, (see 6.4), shall be furnished and installed by the contractor. The rack shall hold the cable and air hose specified herein when the cable and air hose are not in use. The handle and storage rack shall be removable and, when installed, shall not come in contact with any hot components of the compressor.

3.5 Air hose. The air hose shall be fitted at one end with a commercial tire-inflating chuck in lieu of the inflator gage specified in MIL-C-13874.

3.6 Tire gage. A tire pressure gage shall be furnished conforming to GG-G-91, type II, class A, style 1 or 2.

3.6.1 Gage holder. A means for clamping, hanging, or otherwise securing the tire gage to the compressor when the gage is not in use shall be provided. The gage holder shall be located near the air hose rack in an accessible location.

3.7 Air pressure regulator. The air-pressure regulator for adjusting the air pressure shall be of the hand-screw type, shall pass 10 cfm of air at 175 psig, and shall provide a minimum adjustable regulated air-pressure range of from 0 to 175 psig. The regulator shall have anticreepage characteristics to permit a regulated air pressure to be maintained with not more than a 2-psig pressure increase when the regulator is subjected to an inlet supply of 200 psig for a period of 5 minutes with no air flowing through the inflator hose. The regulator shall withstand, without damage, inlet supply pressures up to and including 200 psig. The body of the regulator shall be made of brass, bronze, zinc, aluminum, or other corrosion-resistant metal.

3.7.1 Air pressure gages. The air receiver and the air pressure regulator shall each be equipped with a 0- to 250-psi gage. The gage inlet shall be 1/4-inch male NPT. The case shall be steel, zinc coated or phosphate treated, black enamel finished. The dial shall be not less than 2 inches in diameter. The gages shall be shock mounted.

3.8 Dehydrator. A dehydrator shall be furnished and installed by the contractor. Unless otherwise specified (see 6.2), the dehydrator shall be of the nonregenerative type.

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3.8.1 Dehydrator capacity. While the compressor is operating in ambient conditions of plus 120° F plus or minus 5° F and a relative humidity of 36 percent (180 grains of water per pound of air), the dehydrator shall dry not less than 5 cfm of free air for a period of not less than 6 hours to a level not in excess of minus 40° F free air dew point. During the 6-hour period, no change of dessicant shall be required.

3.8.2 Moisture separator. The dehydrator-inlet shall be equipped with a moisture separator. The moisture separator shall prevent oil, water, and other foreign matter from entering the dehydrator. A manually operated drain valve shall be provided and shall not require operation more often than once every 1/2 hour at a relative humidity of not less than 36 percent at plus 120° F plus or minus 5° F. The filter shall be provided with a thermostatically controlled heater element to prevent the freezing of accumulated moisture at ambient temperatures of minus 25° F to plus 32° F. The heater element shall be rated at 115/230-volts, 60 Hertz, alternating current. The air pressure drop through the moisture separator shall be not more than 10 psig.

3.9 Sound level. The sound level of the compressor while it is operating at full speed, shall be in accordance with MIL-STD-1474, Category D for steady-state-noise when tested in accordance with 4.5.3. If the sound level of the compressor at the microphone location and conditions exceeds 85 dB(A), the distance from the compressor at which a limit of 85 dB(A) is met shall be determined. That distance shall be shown on hazard signs mounted on both sides of each production compressor, stating: "Caution, hearing protection required within ____feet when equipment is operating." The caution shall be legible from the distance shown on the hazard sign. Hearing protectors need not be supplied with the compressor. When the caution notice is required, manuals shall include the appropriate discussion of the noise hazard in accordance with MIL-STD-1474, including the hearing protection, the noise level of the item, and the distance from the compressor at which the 85 dB(A) limit is met.

3.10 Dissimilar Metals. Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other unless the metals are protected against galvanic corrosion (see 6.7).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection comparison (see 4.6).
(d) Inspection of packaging (see 4.7).

4.3 Preproduction inspection. The preproduction compressor(s) shall be tested as specified in 4.5 in addition to the examinations and tests specified in MIL-C-13874.

4.4 Quality conformance inspection. Each production compressor shall be subjected to quality conformance inspection as specified in MIL-C-13874.

4.5 Tests.

4.5.1 Dehydrator.

4.5.1.1 Dehydrator capacity. The compressor shall be operated at an ambient temperature of 120° F plus or minus 5° F and a relative humidity of not less than 36 percent for a period of not less than 6 hours. The dehydrator-inlet moisture separator shall be drained at intervals of not less than 1/2 hour. The following data shall be taken at 1-hour intervals:

(a) Compressor crankshaft speed, in revolutions per minute.
(b) Discharge pressure measured at the dehydrator outlet, in psig.
(c) Ambient air temperature in degrees Fahrenheit, humidity in percent, and pressure in inches of mercury.
(d) Dryness of air produced in degrees Fahrenheit, free air dew point. The dryness should be measured with a hygrometer of the dial indicating, coulometric, electrolytic type.

4.5.1.1.1 Failure criteria. One or more of the following shall constitute failure of this test:

(a) Compressor crankshaft speed in excess of contractor's recommended continuous operating speed.
(b) Discharge pressure less than 200 psig.
(c) Inability of the dehydrator to dry 5 cfm of free air to the dryness specified herein at the conditions specified.

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4.5.1.2 Dehydrator. Two 200 psig test gages shall be installed on the moisture separator, one on the inlet side and one on the outlet side. With the compressor operating, the air receiver discharge valve shall be opened until an air pressure of 175 psig is maintained. The difference in indicated pressure on the inlet and outlet gages shall then be recorded. A pressure drop of more than 10 psig shall constitute failure of this test.

4.5.1.2.1 Valve and heater. The compressor shall be operated at a temperature of minus 25° F plus or minus 5° F for a period of not less than 6 hours. At the end of this period, the drain valve shall be opened. Failure of moisture or air to be discharged or inability of the valve to operate shall constitute failure of this test.

4.5.2 Regulator.

4.5.2.1 Leakage. With the air receiver pressure at 200 psig and the air hose and chuck connected to the regulator, the air receiver discharge valve shall be opened and the air pressure shall be adjusted to 175 psig. Any leakage of air from the chuck, hose, or regulator shall constitute failure of this test.

4.5.2.2 Regulator creep. The air-pressure regulator outlet shall be connected so as to discharge into another air receiver (the air receiver tank of a similar unit may be used, with the air hose as the connecting medium). An air pressure of not less than 200 psig shall be maintained on the air receiver of the compressor being tested. The pressure regulator shall be adjusted to 20 psig delivery pressure for a 5-minute period. An increase in pressure of more than 2 psig at the end of the 5-minute period shall constitute failure of this test. The test shall be repeated for 75 and 150 psig.

4.5.3 Sound level. Place the compressor in a test area which is free of any large reflecting surfaces within 50 feet of the compressor or sound-level measuring equipment. With the compressor operated at rated load, a sound level survey shall be made around the compressor to determine the direction in which the maximum sound levels are propagated so that the test equipment can be properly oriented. The recording microphones shall be positioned in accordance with MIL-STD-1474, Category D. The noise level shall be measured with the compressor operated at no load and rated load. Failure to meet the requirements of 3.9 shall constitute failure of this test.

4.6 Inspection comparison. The Government may select compressors at any time during the contract production period and subject these compressors to the examinations and tests specified in 4.5 or MIL-C-13874 for inspection comparison to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection camparison.

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4.6.1 Inspection failure. Failure of an inspection comparison compressor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison compressor(s) and shall be cause for refusal by the Government to continue acceptance of production compressors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on compressors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all compressors accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. The preservation, packing and marking for level A or level B shall be examined and tested to determine compliance with the quality assurance provisions of MIL-C-3600. The preservation, packing and marking for commercial shall be examined to determine compliance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B or Commercial as specified (see 6.2).

5.1.1 Level A. Each complete compressor with accessories and publications shall be preserved in accordance with the level A preservation requirements of MIL-C-3600.

5.1.2 Level B. Each complete compressor with accessories and publications shall be preserved in accordance with the level B preservation requirements of MIL-C-3600.

5.1.3 Commercial. Each complete compressor with accessories and publications shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. Compressors preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-C-3600.

5.2.2 Level B. Compressors preserved as specified in 5.1, shall be packed in accordance with the level B packing requirements of MIL-C-3600.

5.2.3 Commercial. Compressors preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The compressor is intended to be used at a missile assembly area to test the missile propulsion plumbing for leakage.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
(c) Time frame required for submission of preproduction model and number of compressors to be furnished (see 3.2.1).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2.1).
(e) When cable ends are to be terminated (see 3.3).
(f) When a dehydrator other than as specified is required (see 3.8).
(g) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production compressors from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish compressors conforming to this specification. The contractor should provide a written report to the contracting officer covering all deficiencies, malfunctions, and test failures. This report shall be presented within 48 hours from the time of failure and should describe components and parts affected, both directly and indirectly, corrective action taken or to be taken by the contractor, and the contractor's proposed method of proving the adequacy of the corrective action.

6.4 Information figure. Figure 1 shows a compressor which has been found acceptable; however, the figure is not intended to preclude the furnishing of other compressors which conform to this specification.

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6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each compressor.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of compressors.

6.7 Materiel deterioration prevention and control (MADPAC). The contracting officer should consider for inclusion in all applicable invitations for bid, requests for proposal, and contracts the following statement:

"The technical proposal shall provide assurance that all system/item parts shall be fabricated from compatible materials, suitably finished, to provide protection from corrosion and other forms of materiel deterioration. The assurance shall be in the form of a finish plan, utilizing state-of-art technology, which clearly identifies the specific finish or treatment to be used on the various components and assemblies, to provide protection against the various forms of materiel deterioration in the applicable operating and storage environments to which they may be exposed."

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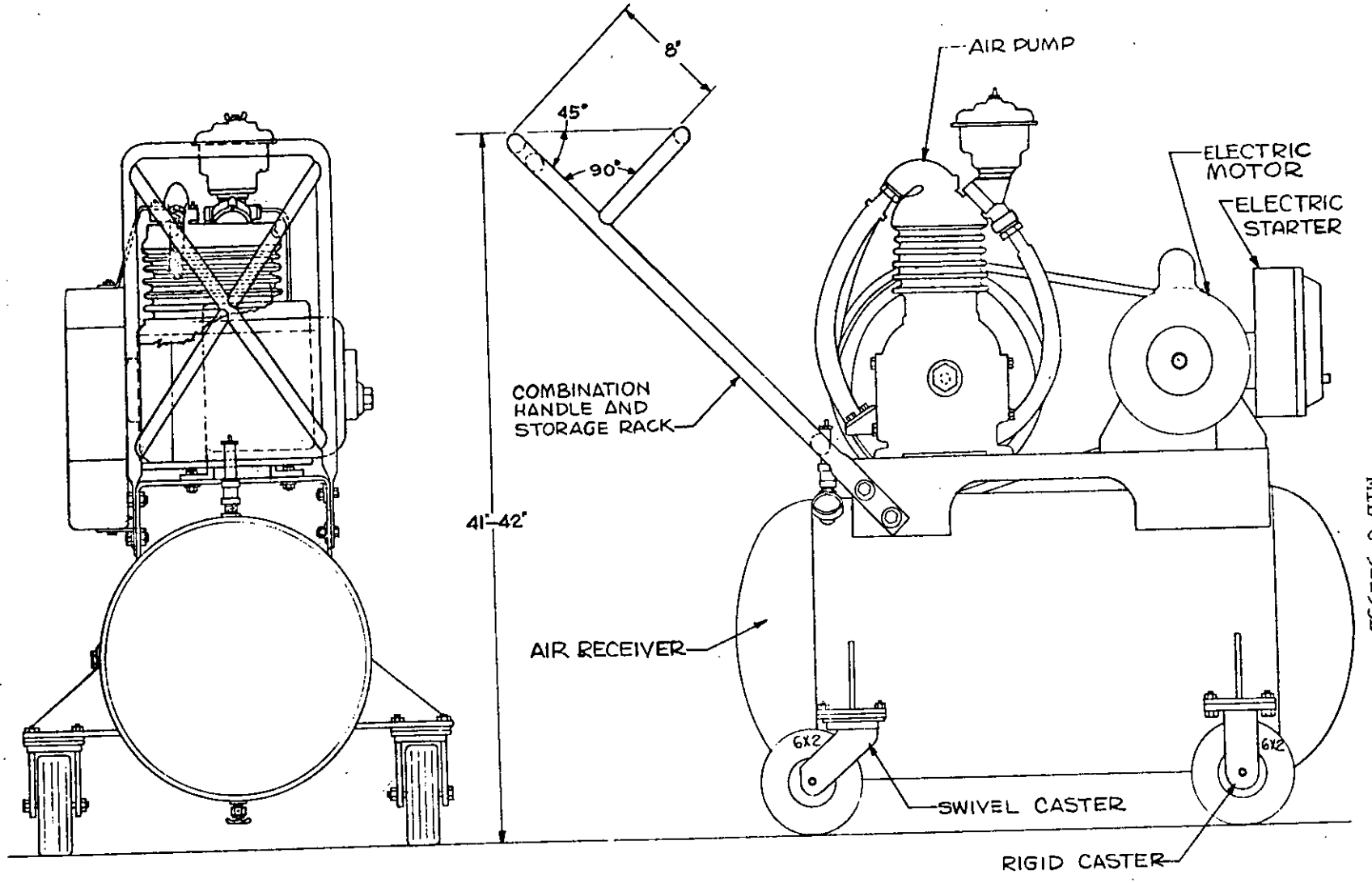


FIG 1 COMPRESSOR, AIR, ELECTRIC MOTOR DRIVEN, 5 CFM, 175 PSIG, CASTER MOUNTED

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-52153B Compressor Unit, Reciprocating, Power-Driven; Air; 5-CFM, 175-PSI, for Testing Missile Plumbing System

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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